

Ferrocene & Sandwich Compounds

Reading Class Quiz — Topic 38

Do this without notes. It is fine to leave boxes blank — you do not have to use every box.

Chain 1 — Aromaticity of the Cyclopentadienyl Anion

Cyclopentadiene (C_5H_6) is not aromatic.



aromatic compounds must be fully conjugated, planar, have $4n+2$ π electrons



cyclopentadiene can lose a hydrogen from the CH_2 group to form the cyclopentadienyl anion, which satisfies those requirements



The cyclopentadienyl anion (Cp^-) is aromatic: it has 6 π -electrons.

Chain 2 — The 18-Electron Rule

The bonding valence orbitals of a first-row transition metal are the $(n-1)d$, ns , and np orbitals (for iron: 3d, 4s, and 4p).



this means that there are 5 d orbitals, 3 p orbitals, and 1 s orbital, making 9 orbitals in total that can hold 18 electrons



filling all valence orbitals achieves a noble gas electron configuration, which is stable because



Stable transition-metal complexes tend to have 18 electrons in their valence bonding orbitals.

Chain 3 — Catalysis

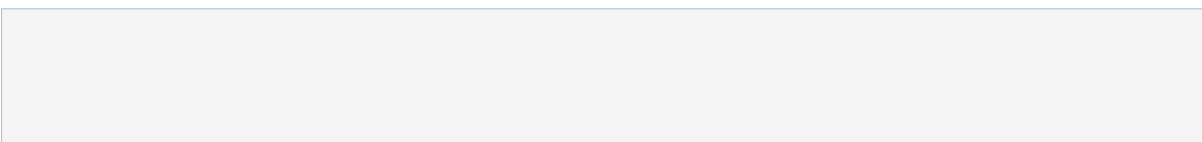
Ferrocene $[\text{Fe}(\eta^5\text{-Cp})_2]$ obeys the 18-electron rule: Fe^{2+} (6 e^-) + $2 \times \text{Cp}^-$ ($2 \times 6\text{ e}^-$) = 18 electrons. There are no vacant coordination sites at iron.



by breaking some of the dative bonds from Cp^- to be less than eta 5, there can be vacant coordination sites at iron



this means that iron can accept and transfer electrons



Ferrocene can still be used as a catalyst, despite seemingly no vacant sites.